ACL Principles and Configuration

Foreword

- Rapid network development brings challenges to network security and quality of service (QoS). Access control lists (ACLs) are closely related to network security and QoS.
- By accurately identifying packet flows on a network and working with other technologies, ACLs can control network access behaviors, prevent network attacks, and improve network bandwidth utilization, thereby ensuring network environment security and QoS reliability.
- This course describes the basic principles and functions of ACLs, types and characteristics of ACLs, basic composition of ACLs, ACL rule ID matching order, usage of wildcards, and ACL configurations.

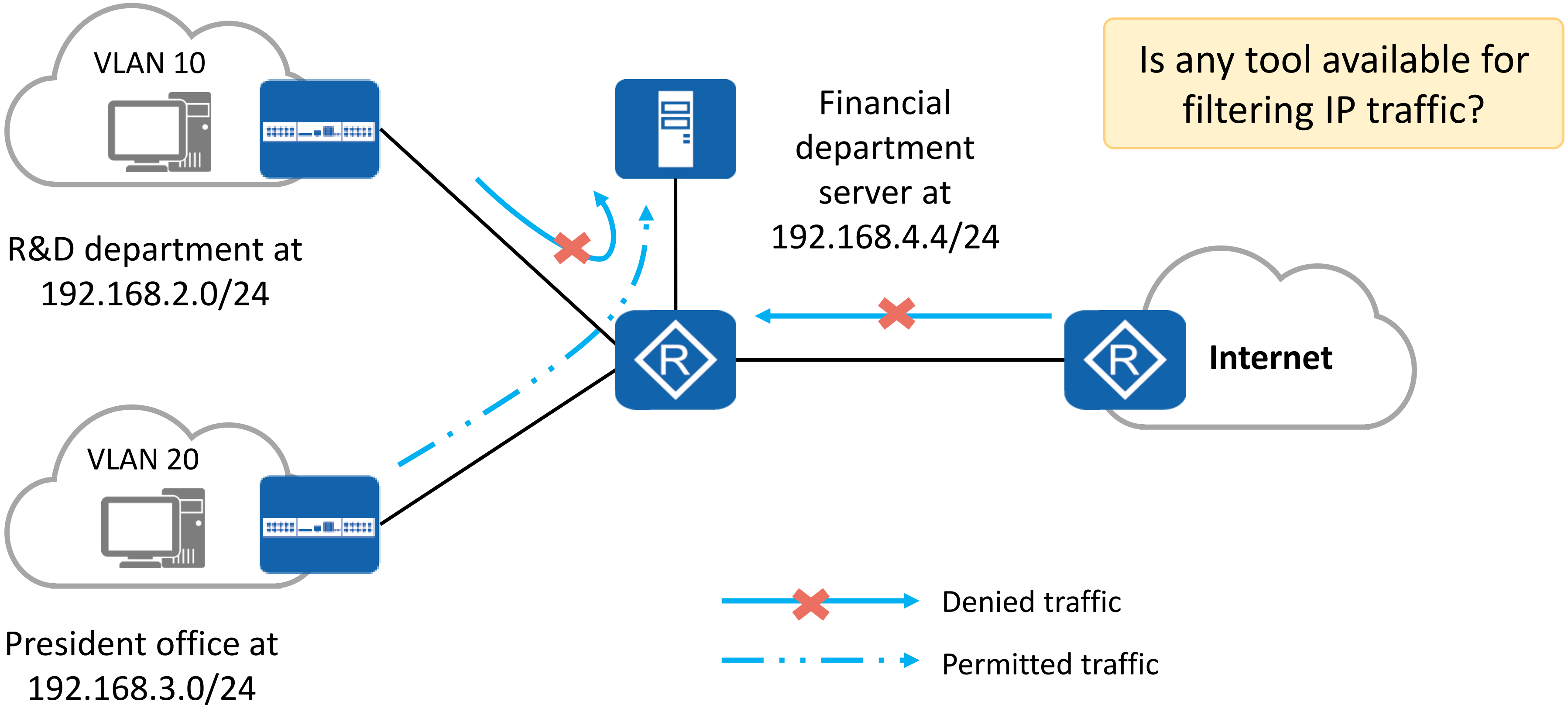
Objectives

- On completion of this course, you will be able to:
  - Describe the basic principles and functions of ACLs.
  - Understand the types and characteristics of ACLs.
  - Describe the basic composition of ACLs and ACL rule ID matching order.
  - Understand how to use wildcards in ACLs.
  - Complete the basic configurations of ACLs.

Contents

1. **ACL Overview**
2. Basic Concepts and Working Mechanism of ACLs
3. Basic Configurations and Applications of ACLs

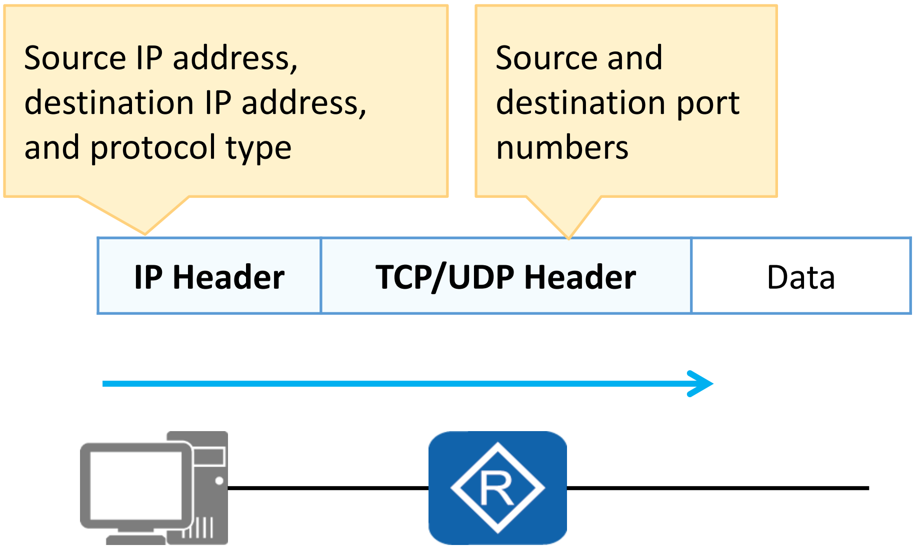
Background: A Tool Is Required to Filter Traffic



- To ensure financial data security, an enterprise prohibits the R&D department's access to the financial department server but allows the president office's access to the financial department server.

ACL Overview

- An ACL is a set of sequential rules composed of permit or deny statements.
- An ACL matches and distinguishes packets.



ACL Application

- Matching IP traffic
- Invoked in a traffic filter
- Invoked in network address translation (NAT)
- Invoked in a routing policy
- Invoked in a firewall policy
- Invoked in QoS
- Others

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ACL Composition

- An ACL consists of several permit or deny statements. Each statement is a rule of the ACL, and permit or deny in each statement is the action corresponding to the rule.

What does each rule mean?

acl number 2000 — ACL number

```
rule 5   permit   source 1.1.1.0 0.0.0.255
rule 10 deny      source 2.2.2.0 0.0.0.255
rule 15 permit   source 3.3.3.0 0.0.0.255
...
```

User-defined rules

Rule ID

Action

Matching option (source IP address)

```
rule 4294967294 deny
```

Rule hidden at the end of the ACL

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Rule ID

```
acl number 2000
                Rule ID
rule     5     deny     source  10.1.1.1  0
rule     10    deny     source  10.1.1.2  0
rule     15    permit   source  10.1.1.0  0.0.0.255
            Step = 5
```

How do I add a rule?

rule 11 deny source 10.1.1.3 0

```
acl number 2000
rule     5     deny     source  10.1.1.1  0
rule     10    deny     source  10.1.1.2  0
rule     11    deny     source  10.1.1.3  0
rule     15    permit   source  10.1.1.0  0.0.0.255
```

Rule ID and Step

- **Rule ID**
  Each rule in an ACL has an ID.
- **Step**
  A step is an increment between neighboring rule IDs automatically allocated by the system. The default step is 5. Setting a step facilitates rule insertion between existing rules of an ACL.
- **Rule ID allocation**
  If a rule is added to an empty ACL but no ID is manually specified for the rule, the system allocates a step value (5 for example) as the ID of the rule. If an ACL contains rules with manually specified IDs and a rule with no manually specified ID is added, the system allocates to this rule an ID that is greater than the largest rule ID in the ACL and is the smallest integer multiple of the step value.

Wildcard (1)

```
acl number 2000
                                                    Wildcard
rule   5    deny    source  10.1.1.1  0
rule   10   deny    source  10.1.1.2  0
rule   15   permit  source  10.1.1.0  0.0.0.255
```

Wildcard

- A wildcard is a 32-bit number that indicates which bits in an IP address need to be strictly matched and which bits do not need to be matched.
- A wildcard is usually expressed in dotted decimal notation, as a network mask is expressed. However, their meanings are different.

- Matching rule

  0: matching; 1: random allocation

How do I match the network segment address corresponding to 192.168.1.1/24?

192.168.1.1	1 1 0 0 0 0 0 0	1 0 1 0 1 0 0 0	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1
0.0.0.255	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	1 1 1 1 1 1 1 1
		Strict matching		Random allocation

192.168.1.0/24 network segment

Wildcard (2)

- A wildcard can be used to match odd IP addresses in the network segment 192.168.1.0/24, such as 192.168.1.1, 192.168.1.3, and 192.168.1.5.

Strict matching   Random allocation   Strict matching

192.168.1	1							
192.168.1	0	0	0	0	0	0	0	1

192.168.1	3							
192.168.1	0	0	0	0	0	0	1	1

192.168.1	5							
192.168.1	0	0	0	0	0	1	0	1

...

Wildcard

0.0.0.	1	1	1	1	1	1	1	0

192.168.1.1 0.0.0.254

The value 1 or 0 in the wildcard can be inconsecutive.

Special Wildcard

- Exactly match the IP address 192.168.1.1.

  192.168.1.1 0.0.0.0 = 192.168.1.1 0

- Match All IP addresses.

  0.0.0.0 255.255.255 = any

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ACL Classification and Identification

- ACL classification based on ACL rule definition methods

Category	Number Range	Description
Basic ACL	2000 to 2999	Defines rules based on source IPv4 addresses, fragmentation information, and effective time ranges.
Advanced ACL	3000 to 3999	Defines rules based on source and destination IPv4 addresses, IPv4 protocol types, ICMP types, TCP source/destination port numbers, UDP source/destination port numbers, and effective time ranges.
Layer 2 ACL	4000 to 4999	Defines rules based on information in Ethernet frame headers of packets, such as source and destination MAC addresses and Layer 2 protocol types.
User-defined ACL	5000 to 5999	Defines rules based on packet headers, offsets, character string masks, and user-defined character strings.
User ACL	6000 to 6999	Defines rules based on source IPv4 addresses or user control list (UCL) groups, destination IPv4 addresses or destination UCL groups, IPv4 protocol types, ICMP types, TCP source/destination port numbers, and UDP source/destination port numbers.

- ACL classification based on ACL identification methods

Category	Description
Numbered ACL	Traditional ACL identification method. A numbered ACL is identified by a number.
Named ACL	A named ACL is identified by a name.

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Basic and Advanced ACLs

- Basic ACL

Number range: 2000 to 2999

Source IP address

IP Header	TCP/UDP Header	Data

```
acl number 2000
 rule    5    deny     source  10.1.1.1  0
 rule    10   deny     source  10.1.1.2  0
 rule    15   permit   source  10.1.1.0  0.0.0.255
```

- Advanced ACL

Number range: 3000-3999

Source IP address, destination IP address, and protocol type

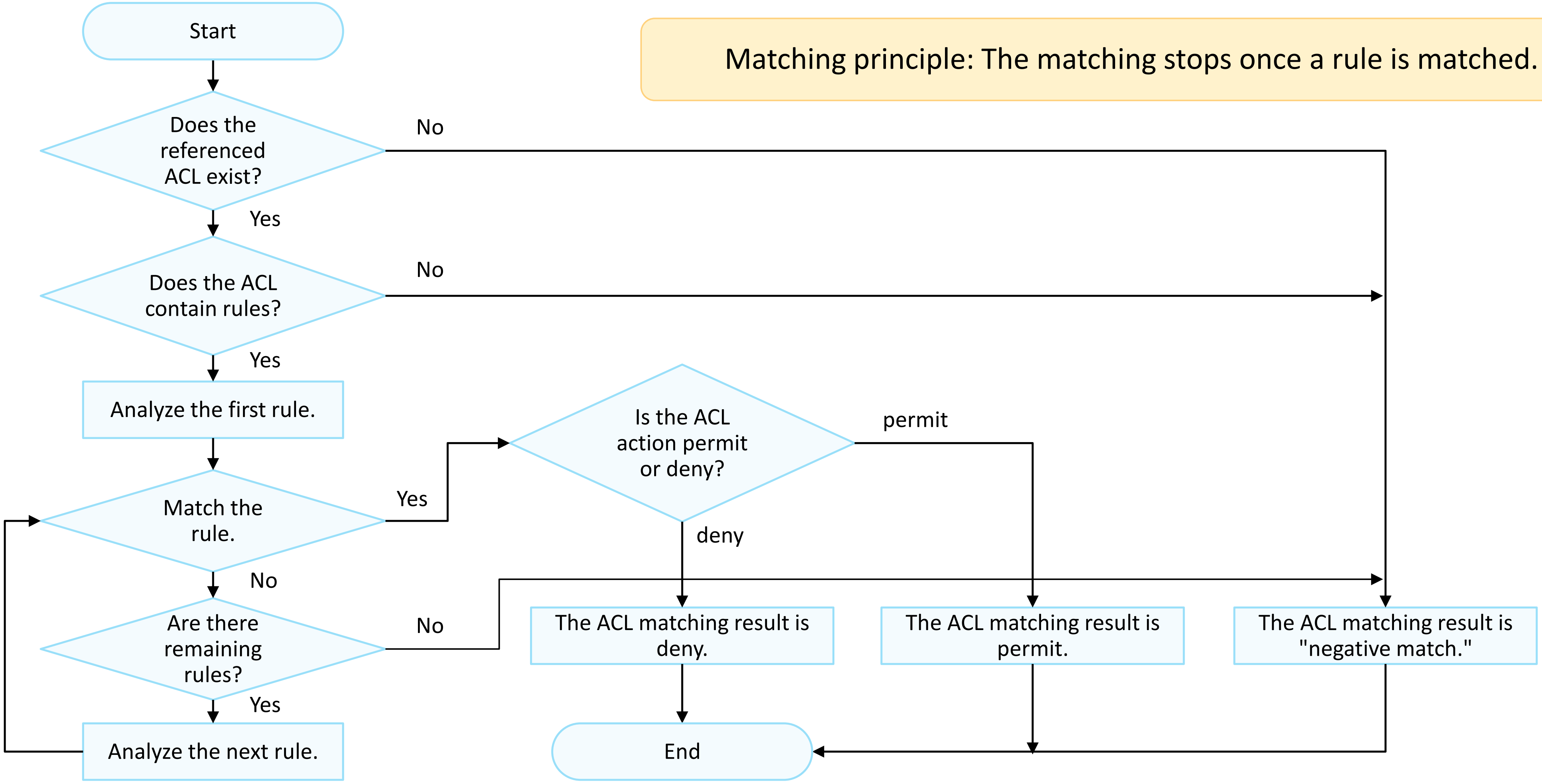
Source and destination port numbers

IP Header	TCP/UDP Header	Data

```
acl number 3000
 rule  5  permit ip       source  10.1.1.0  0.0.0.255  destination  10.1.3.0  0.0.0.255
 rule  10 permit tcp  source  10.1.2.0   0.0.0.255  destination  10.1.3.0  0.0.0.255  destination-port eq 21
```

ACL Matching Mechanism

Matching principle: The matching stops once a rule is matched.



ACL Matching Order and Result

- Configuration order (config mode)
  - The system matches packets against ACL rules in ascending order of rule ID. That is, the rule with the smallest ID is processed first.

Object to be matched

192.168.1.1/24
192.168.1.2/24
192.168.1.3/24
192.168.1.4/24
192.168.1.5/24

Basic ACL

```
acl 2000
  rule 1 permit source 192.168.1.1 0.0.0.0
  rule 2 permit source 192.168.1.2 0.0.0.0
  rule 3 deny    source 192.168.1.3 0.0.0.0
  rule 4 permit 0.0.0.0 255.255.255.255
```

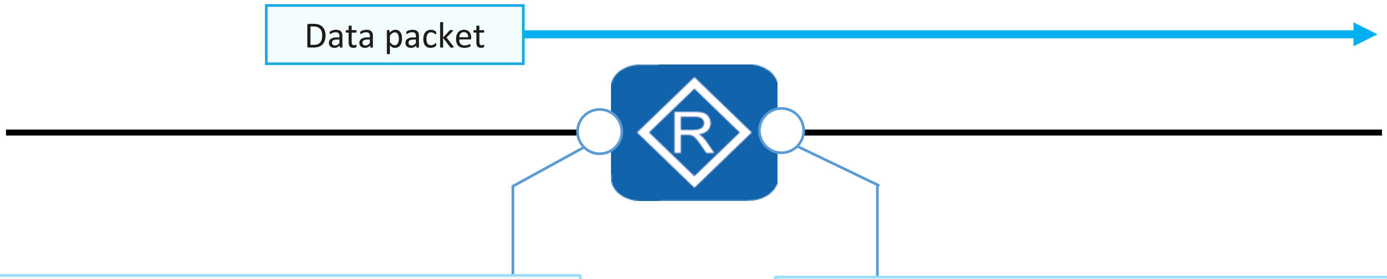
Permitted IP addresses

192.168.1.1/24
192.168.1.2/24
192.168.1.4/24
192.168.1.5/24

rule 1: permits packets with the source IP address 192.168.1.1.
rule 2: permits packets with the source IP address 192.168.1.2.
rule 3: denies packets with the source IP address 192.168.1.3.
rule 4: permits packets from all other IP addresses.

Does "permit" mean that traffic is allowed to pass?

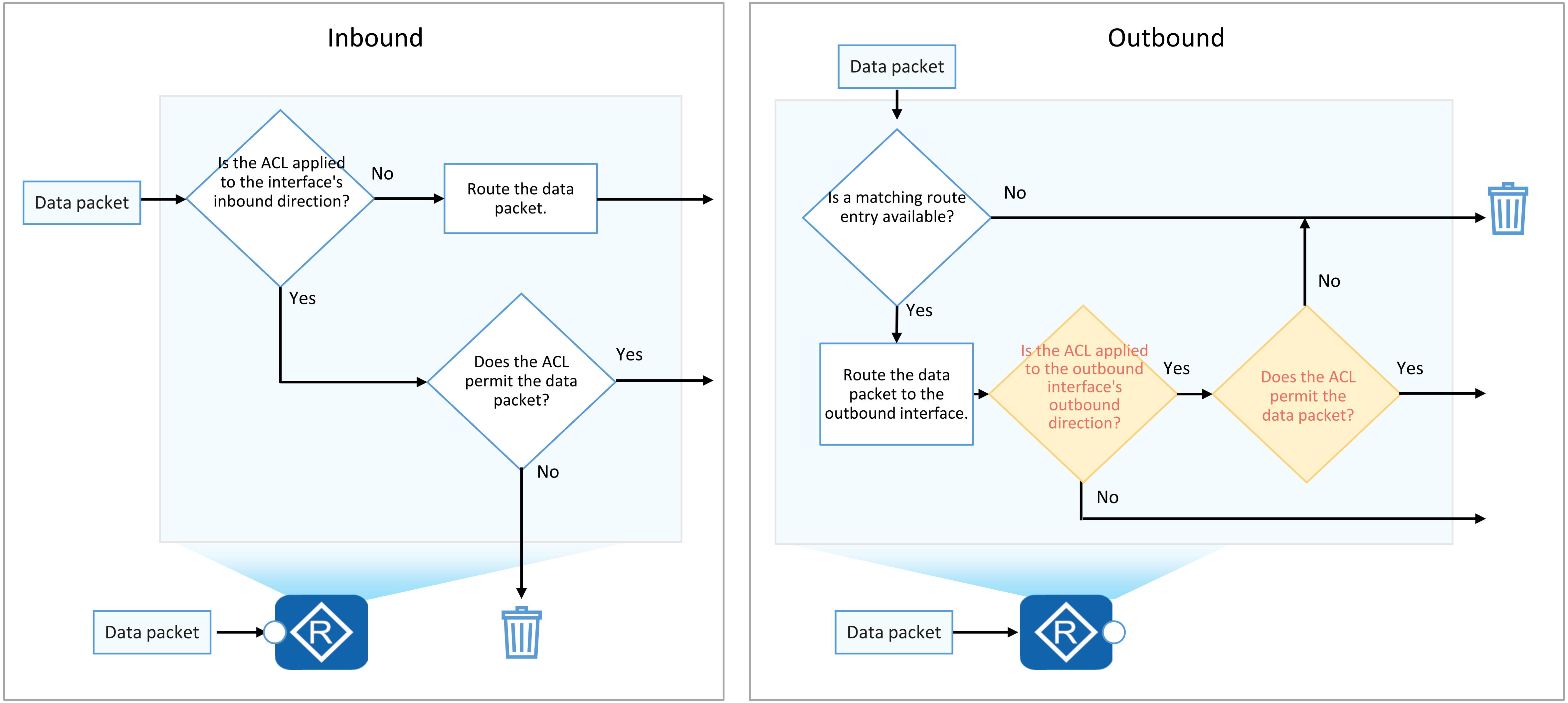
ACL Matching Position



Configure an ACL on the interface.
To enable the ACL to take effect for the data packet shown in the figure,
apply the ACL to the inbound direction.

Configure an ACL on the interface.
To enable the ACL to take effect for the data packet shown in the figure,
apply the ACL to the outbound direction.

Inbound and Outbound Directions



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Basic Configuration Commands of Basic ACLs

1. Create a basic ACL.

[Huawei] **acl** [**number**] *acl-number* [**match-order config**]

Create a numbered basic ACL and enter its view.

[Huawei] **acl name** *acl-name* { **basic** | *acl-number* } [**match-order config**]

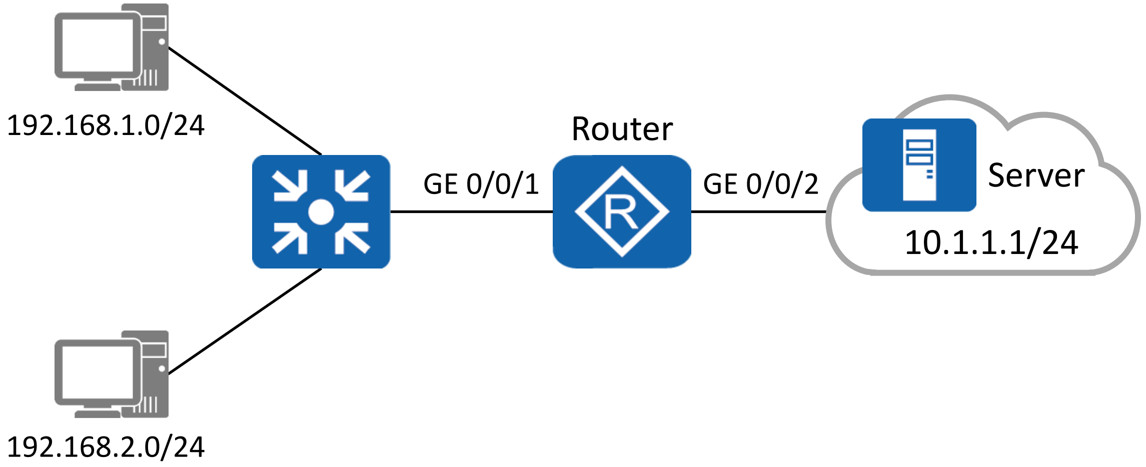
Create a named basic ACL and enter its view.

2. Configure a rule for the basic ACL.

[Huawei-acl-basic-2000] **rule** [*rule-id*] { **deny** | **permit** } [**source** { *source-address source-wildcard* | **any** } | **time-range** *time-name*]

In the basic ACL view, you can run this command to configure a rule for the basic ACL.

Case: Use a Basic ACL to Filter Data Traffic



- Requirements:

  To prevent the user host on the network segment 192.168.1.0/24 from accessing the network where the server resides, configure a basic ACL on the router. After the configuration is complete, the ACL filters out the data packets whose source IP addresses are on the network segment 192.168.1.0/24 and permits other data packets.

1. Configure IP addresses and routes on the router.

2. Create a basic ACL on the router to prevent the network segment 192.168.1.0/24 from accessing the network where the server resides.

```
[Router] acl 2000
[Router-acl-basic-2000] rule deny source 192.168.1.0  0.0.0.255
[Router-acl-basic-2000] rule permit source any
```

3. Configure traffic filtering in the inbound direction of GE 0/0/1.

```
[Router] interface GigabitEthernet 0/0/1
[Router-GigabitEthernet0/0/1] traffic-filter inbound acl 2000
[Router-GigabitEthernet0/0/1] quit
```


Basic Configuration Commands of Advanced ACLs (1)

1. Create an advanced ACL.

```
[Huawei] acl [ number ] acl-number [ match-order config ]
```

Create a numbered advanced ACL and enter its view.

```
[Huawei] acl name acl-name { advance | acl-number } [ match-order config ]
```

Create a named advanced ACL and enter its view.

Basic Configuration Commands of Advanced ACLs (2)

2. Configure a rule for the advanced ACL.
   You can configure advanced ACL rules according to the protocol types of IP packets. The parameters vary according to the protocol types.

   - When the protocol type is IP, the command format is:

   rule [*rule-id*] { **deny** | **permit** } **ip** [**destination** { *destination-address destination-wildcard* | **any** } | **source** { *source-address source-wildcard* | **any** } | **time-range** *time-name* | [**dscp** *dscp* | [**tos** *tos* | **precedence** *precedence*]]]

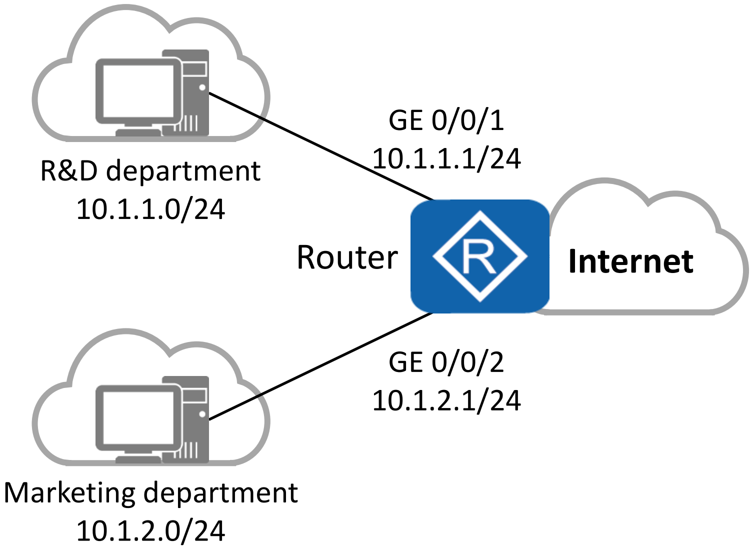
   In the advanced ACL view, you can run this command to configure a rule for the advanced ACL.

   - When the protocol type is TCP, the command format is:

   rule [*rule-id*] { **deny** | **permit** } { *protocol-number* | **tcp** } [**destination** { *destination-address destination-wildcard* | **any** } | **destination-port** { **eq** port | **gt** port | **lt** port | **range** port-start port-end } | **source** { *source-address source-wildcard* | **any** } | **source-port** { **eq** port | **gt** port | **lt** port | **range** port-start port-end } **|** **tcp-flag** { **ack** | **fin** | **syn** } * | **time-range** time-name] *

   In the advanced ACL view, you can run this command to configure a rule for the advanced ACL.

Case: Use Advanced ACLs to Prevent User Hosts on Different Network Segments from Communicating (1)



Requirements:

- The departments of a company are connected through the router. To facilitate network management, the administrator allocates IP addresses of different network segments to the R&D and marketing departments.
- The company requires that the router prevent the user hosts on different network segments from communicating to ensure information security.

1. Configure IP addresses and routes on the router.

2. Create ACL 3001 and configure rules for the ACL to deny packets from the R&D department to the marketing department.

[Router] **acl** 3001

[Router-acl-adv-3001] **rule deny ip source** 10.1.1.0 0.0.0.255 **destination** 10.1.2.0 0.0.0.255

[Router-acl-adv-3001] quit

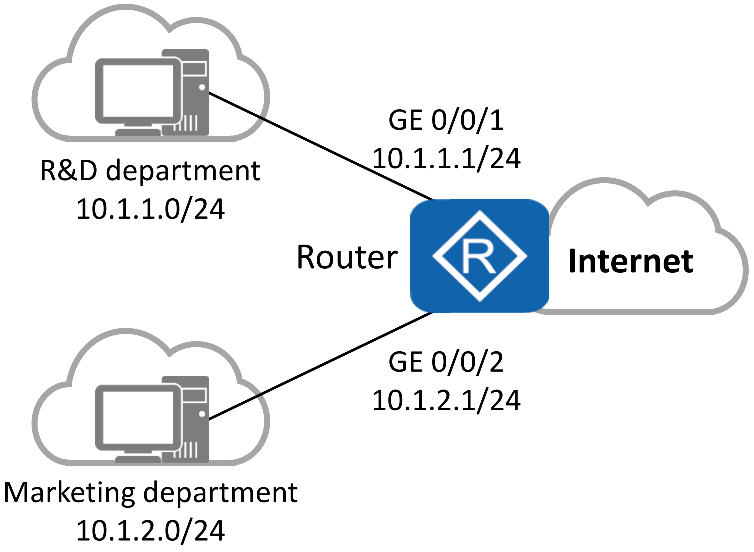
3. Create ACL 3002 and configure rules for the ACL to deny packets from the marketing department to the R&D department.

[Router] **acl** 3002

[Router-acl-adv-3002] **rule deny ip source** 10.1.2.0 0.0.0.255 **destination** 10.1.1.0 0.0.0.255

[Router-acl-adv-3002] quit

Case: Use Advanced ACLs to Prevent User Hosts on Different Network Segments from Communicating (2)



Requirements:

- The departments of a company are connected through the router. To facilitate network management, the administrator allocates IP addresses of different network segments to the R&D and marketing departments.
- The company requires that the router prevent the user hosts on different network segments from communicating to ensure information security.

4. Configure traffic filtering in the inbound direction of GE 0/0/1 and GE 0/0/2.

```
[Router] interface GigabitEthernet 0/0/1
[Router-GigabitEthernet0/0/1] traffic-filter inbound acl 3001
[Router-GigabitEthernet0/0/1] quit

[Router] interface GigabitEthernet 0/0/2
[Router-GigabitEthernet0/0/2] traffic-filter inbound acl 3002
[Router-GigabitEthernet0/0/2] quit
```

Quiz

1. (Single) Which one of the following rules is a valid basic ACL rule? ( )

    A. rule permit ip

    B. rule deny ip

    C. rule permit source any

    D. rule deny tcp source any

2. Which parameters can you use to define advanced ACL rules?

Summary

- ACL is a widely used network technology. Its principle is as follows: packets are matched against configured ACL rules and actions are taken on the packets as configured in the ACL rules. The matching rules and actions are configured based on network requirements. Due to the variety of matching rules and actions, ACLs can implement a lot of functions.
- ACLs are often used with other technologies, such as firewall, routing policy, QoS, and traffic filtering.

Thank You

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